CADT IDT
វិទ្យាស្ថានបច្ចេកវិទ្យាឌីជីថល
Institute of Digital Technology

Computer Architecture Lab (Week 5)

CADT IDT វិទ្យាស្ថានបច្ចេកវិទ្យាឌីជីថល Institute of Digital Technology

TP01: Write an assembly language to calculate (a + b)^2 and store answer in location (2005)

Sample Problem:

- Store Let a = 09 in memory location 2000H and b = 0A in memory location 2001H
- Calculate (a + b)^2 = (13)^2 = 0169
- Store 69H to memory location 2005H and 01H in memory location 2006H

TP02: Write an assembly language to calculate whether m X 2n and store answer in address 2000

Sample Problem:

- Store Let n = 1E in memory location 2000H and m = 0A in memory location 2001H
- Calculate m X 2n = 0A X 2(1E)= 0258
- Store 58H to memory location 2005H and 02H in memory location 2006H

TP03: Write an assembly language to calculate whether (a+b)^2 or (a-b)^2

Sample Problem:

- Store Let a = 04H, b = 07H, c (status) = 02H
- Calculate (a – b) ^ 2 = (04-07)^2= 09H
- Store 09H to memory location 2005H

TP04: Write an assembly language to calculate (a+b)^2 -2b where (a+b)^2 > 2b

- Let a in location @2000H, b in location @2001H
- Result store in location 2005H

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TP05: Write an assembly language to calculate (a-b)^2 +3c

- Let a in location @2000H, b in location @2001H, and c in location @2002H
- Result store in location 2005H

TP06: Write an assembly language to calculate $(a-b)^2 + 4c$

- Let a in location @2000H, b in location @2001H, and c in location @2002H
- Result store in location 2005H

TP07: Write an assembly language to calculate the division of two 8-bit numbers. (a>b)

Sample Problem:

- Store Let a = 1FH, b = 05H
- Calculate a/b = = 06H, and remain = 01H
- Store 06H to memory location 2005H and remain 01H in memory location 2009

TP08: Write an assembly language to find minimum value among 5 numbers

Sample Problem:

- All 5 number store in memory location from 2000H to 2004H
- Length of series store in memory location 2005H
- Store the smallest number in memory location 2010H

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TP09: Write an assembly language to find maximum value among 7 numbers.

Sample Problem:

- All 7 number store in memory location from 2000H to 2006H
- Length of series store in memory location 2007H
- Store the biggest number in memory location 2010H

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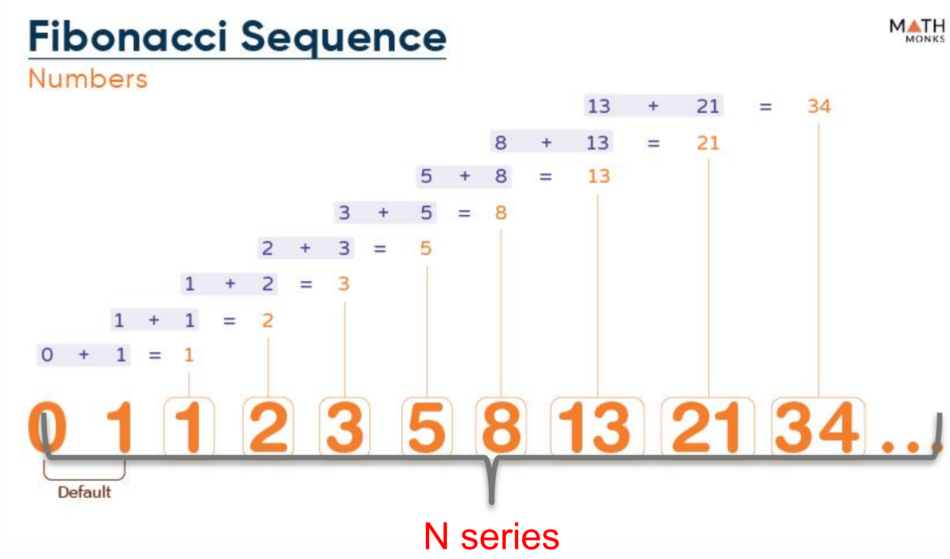
TP010: Write an assembly language to display Fibonacci numbers.

Store in any range of memory location. Ex N = 5 store from memory location @2000 to @2004.



Thanks